Example 7.1:-

The radiative and nonradiative recombination lifetimes of the minority carriers in the active region of a double-heterojunction LED are 60ns and 100ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87 µm at a drive current of 40mA.

Solution:-

The total carrier recombination lifetime:-

$$\tau = \frac{\tau_r \tau_{nr}}{\tau_r + \tau_{nr}} = \frac{60*100 ns}{60+100} = 37.5 ns.$$

$$\eta_{int} = \frac{\tau}{\tau_r} = \frac{37.5}{60} = 0.625$$

$$\therefore P_{int} = \eta_{int} * \frac{hci}{e\lambda}$$

$$= \frac{0.625 \times 6.626 \times 10^{-34} \times 2.998 \times 10^{8} \times 40 \times 10^{-3}}{1.602 \times 10^{-19} \times 0.87 \times 10^{-6}}$$

$$= 35.6 mW.$$

(7.2) A planar LED is fabricated from gallium arsenide which has a refractive index of 3.6.

(a) Calculate the optical powder emitted into air as a ~~presen~~ percentage of the internal optical powder for the device when the transmission factor at the crystal–air interface is 0.68.

(b) When the optical powder generated internally is 50% of the electric powder supplied, determined the external power efficiency.

(a) Ans:-

$$P_e \cong \frac{P_{int} F n^2}{4 n_x^2} = \frac{P_{int}\, 0.68 * 1}{4(3.6)^2} = 0.013 P_{int}.$$

(b) Ans:-

$$\eta_{ep} = \frac{P_e}{P} * 100 = 0.013 * \frac{P_{int}}{P} * 100$$

Also, the optical power generated internally

$$P_{int} = 0.5P$$

$$\eta_{ep} = \frac{0.013 * P_{int}}{2 P_{int}} * 100 = 0.65\%.$$

Example 7·3:- The light output from the GaAs LED for example 7·2 is coupled into a step index fiber with a numerical aperture of 0·2, a core refractive index of 1·4 a diameter larger than the diameter of the device. Estimate:

(a) The coupling efficiency into the fiber when the LED is in close proximity to fiber core.

(b) The optical loss in decibles, relative to the power emitted from the LED, when coupling the light output into the fiber.

(c) The loss relative to the internally generated optical power in the device when coupling the light output into the fiber when there is a small airgap between the LED and the fiber core.

Solution:-

(a) $\eta_e = (NA)^2 = (0{\cdot}2)^2 = 0{\cdot}04$

Thus' about 4% of the externally emitted optical power is coupled into the fiber.

(b) Ans:-

Let the optical power coupled into the fiber be P_c. Then the ~~couple~~ optical loss in decibels relative to P_o when coupling the light output into the fiber is:—

$$Loss = -10 \log_{10} \frac{P_c}{P_o}$$

$$= -10 \log_{10} \eta_c$$

Hence,

$$Loss = -10 \log_{10} 0.04$$

(c) When the LED is emitting into air,

$P_e = 0.013 P_{int}$.

Assuming a very small air grip, then from (a) powerful coupled into the fiber is:—

$$P_c = 0.04 P_e = 0.04 * 0.013 P_{int}$$

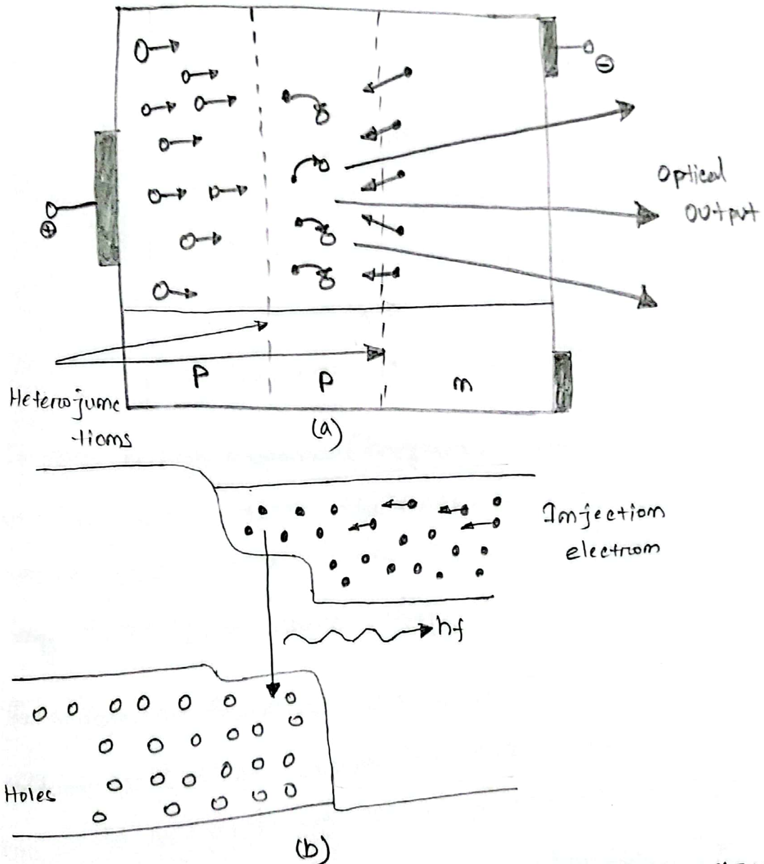
$$= 5.2 * 10^{-4} P_{int}.$$

$$Loss = -10 \log_{10} \frac{P_c}{P_{int}} = -10 \log_{10} 5.2 * 10^{-4} = 32.8 dB$$

The double-heterojunction LED:-

The principle of operation of DH LED is illustrated. The device shown consists of a p-type GaAs layer sandwitched between a p-type AlGaAs and an n-type AlGaAs layer. When a forward bias is applied electrons from the n-type layer are injected through the p-n junction into the p-type GaAs layer where they become minority carriers. These minority carriers diffuse away from the junction, recombining with majority carriers (holes) as they do so. Photons are therefore produced with energy corresponding ~~the~~ to the bandgap energy of the p-type GaAs layer.

The injected electrons are inhibited from diffusing into p-type AlGaAs layer because of the potential barrier presented by the p-p heterojunction. Hence, electroluminescence only occurs in the GaAs junction layer, providing both good internal quantum efficiency & high radiance emission. Furthermore, light is emitted from the device without reabsorption because the bandgap energy in the AlGaAs layers is large in



The double-heterojunction LED: (a) the layers structure, shown with an applied forward bias, (b) the corresponding energy band diagram

Surface emitter LEDs:-

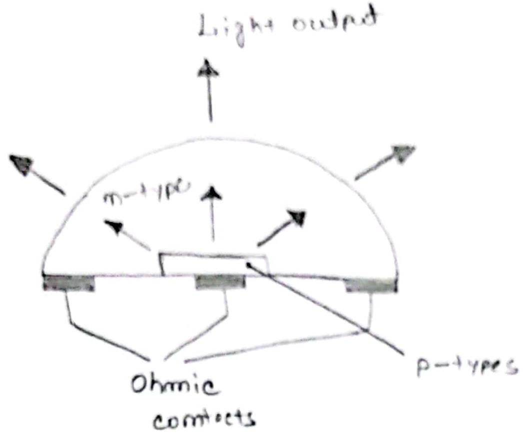


Fig: The structure of the LED dome

To receive high radiance (brightness) in a device, one method is too focus the emission into a small active area. Burrus and Dowson pioneered a teaching technique where they etched a well in a GaAs (Gallium Arsenide) substrate. This design helps reduce radiation absorbtion and also makes it easier to connect an optical fiber to the device. These structures handle heat well, allowing higher currents and brighter emission into the fiber.

Using double heterostructure (DH) designs improves efficiency because they confine both the electrical current and light, reducing energy loss. This type of device, called a surface emitter LED (SLED), is commonly used in optical fiber communications especially for the 0.8 to 0.9 μm wavelength range. The design reduces internal absorption and reflects more ~~to~~ light forward, making it brighter.

The light emitted is spread out evenly (isotropic) but when it exists the device & enters the fiber the distribution is shaped more like a wide-angle beam due to the change in refractive index.

The power coupled into a fiber from the SLED can be estimated using a formula, where the key factors include the reflection at the fiber surface, the size of the fiber core, and the brightness of the SLED. However, the actual

Power depends on other practical details like the distance and alignment between the device & fiber, and the material used.

Edge emitter LED:-

Edge emitter LED is a high-radiance device used in optical communications, similar in design to a strip geometry laser. It has a very thin active layer (50 to 100 μm) that allows light to spread into surrounding transparent layers, reducing absorption. This creates a waveguid that narrows the beam's spread to about 30° in one direction, the light spreads widely (120°), similar to a surface emitter.

Most light is emitted from one end, thanks to a reflector on the opposit end and an anti-reflection coating on the emitting side. ELEDs can couple more power into small numerical aperture (NA) fibers compared to ~~small~~ surface emitters, but surface emitters generally radiate more total power into the air.

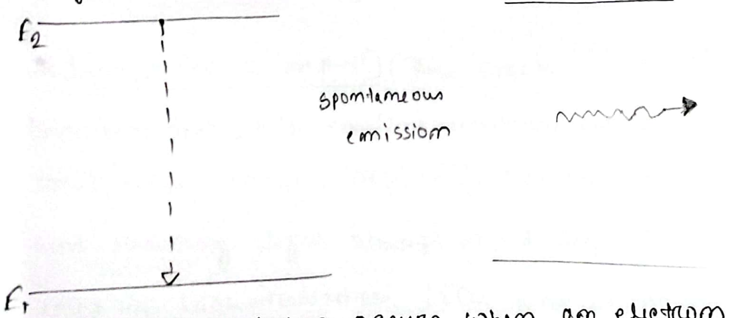
Edge emitters work better for small NA (less than 0.3) with surface emitter perform better with large NA fibers.

Spontaneous Emission:-

* For most of the light sources the photons are emitted spontaneously.

* In a first step, an electron is elevated to a energetically higher state which is very unstable. In second step, the electron will spontaneously return to an energetically more stable state.

* This process is a statistical process which can happen very fast. As a consequence the spontaneously emitted photons are incoherent and the emitted spectrum has broad spectral width. Energy state diagram for spontaneous emission of a photon.



* Spontaneous emission occurs when an electron that is in an excited state spontaneously transmissions to a lower energy state.
* When this occurs, a photon is emitted. The energy of the photon is equal to the energy difference between

the two energy levels.

(2) Hetero-junctions:-

A homojunction is a p-n junction made from a single semiconductor material. However, using heterojunctions - interferences between two different semiconductors with varying bandgap energies - can enhance the radiative properties of devices like LEDs and Lasers. Devices with heterojunctions are called heterostructures.

There are two types of heterojunctions:-

* Isotype (n-n or p-P): These create potential barriers that help confine minority carriers in a small active region improving radiative recombination and reducing light absorption. They are commonly used in high-radiance LEDs and lasers.

* Anisotype (p-n):- These improve the injection efficiency of electrons or holes, and both types

help confine radiation to the active region by creating a ~~diee~~ dielectric step.

Double heterojunctions (DH) are especially useful in devices like injection ~~lasers~~ lasers. They confine both carriers and light to the active region, which dramatically reduces the current needed to achive lasing, making the device much more efficient. In DH lasers, careful fabrication ensures that refractive index steps at the heterojunctions confine the emitted well, acting as a waveguid.

For efficient performance, the material used in hetero junctions should have closedly match lattice by causing non-radioactive recombination. Although, perfect lattice matching isn't always possible, a missmatch up around 0.6% is often tolerated.

Distributed Feedback Lasers:-

* Used to generate signal with signal frequency and less susceptible to temperature variation.

* It has diffraction gramting above region and diffraction grating fraction as bragg reflecton.

* Distributed Bragg diffraction grating provides periodic variation in refractive index.

* Feedback of optical energy is obtained through bragg reflecton rather than by usual cleaved mirrors.

* An optical bragg grating is transparent device with a periodic variation of the refractive index, so that a large Reflectance (reflectively) may be reached in some wavelength range (bandwidth) around a certain wavelength which fullfils the brag condition! The angle of incidence = angle of scattering.

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* Single-frequency operation of (a) the distributed feedback (DFB) laser in comparison with (b) the Fabry-Perot ~~las~~ laser.

* When the period of corrugation is equal to $l\lambda_B/2n_e$. when l is the integer order of the grating, λ_B = is the Bragg wavelength

n_e = effective refractive index of the waveguide

Then only the mode near the Bragg wavelength λ_B is reflected constructively.

* From the viewpoint of device operation, semiconductor lasers employing the distributed feedback mechanism can be said,

(a) Distributed feedback (DFB) laser

(b) Distributed Bragg reflector (DBR) laser

* A distributed feedback laser replaces the back mirror with a grating along ~~with~~ the cavity axis.

* Since the light is being reflected by the grating the reflected light is always in the correct phase no matter if it was reflected from the front or back of the grating.

* The distributed nature of the Reflection sharpens the cavity resonance & distributed feedback lasers are typically of much narrow bandwidth than the same laser with mirrors.

* DBR laser the grating is etched only near the cavity ends and hence distributed feedback does not occur in the central active region.

* Both DBR & DFB lasers are used to provide single-frequency semiconductor optical sources.

Short- & coupled- cavity lasers:-

Short cavity lasers:- One way to improve the selection of a single longitudinal mode in an injection ~~for~~ laser is to shorten the cavity lengths. Reducing the cavity from 250µm to around 25µm increases the spacing between modes from 1 to 10 nm, making it easier to select a single method. However, fabricating very short microcleaved and etched resonators are used. These are create short cavities (10 to 20 µm) perpendicular to the active region, enabling stable single-frequency operation.

Coupled-cavity lasers offer another method for single frequency operation. In these lasers, multiple resonators ~~&~~ or distributed reflectors introduce frequency- dependent losses, allowing single-mode oscillation when the longitudinal modes of each cavity align. An example is the three-mirror resonator, which uses a graded-index (GRIN) lense for coupling.

Another approach is the cleaved-coupled-cavity (C^3) laser, where two laser sections are seperated by a small gap (around one wavelength). This creates a four-mirror resonator that can achive strong single-mode operation and high side-mode suppression by controlling the injection currents and temperature. Additionaly, C^3 lasers are tunable, with mode jumps around 2nm, allowing for discrete tuning over a 26 nm range by adjusting the current one section.